| | |
|---|---|
| **EQUIPMENT NAME:** | **FIM III Booster Station 33/11kV Transformer** |
| **EQUIPMENT NUMBER:** | **F23-TR-601** |

| **Procurement Package** | | "EL007" - 33/11 kV Fluid Filled Transformer | | | | | |
|---|---|---|---|---|---|---|---|
| **Equipment No.** | **Rating (kVA)** | **Voltage Ratio (kV)** | **Vector Group** | **Type** | **Insulation** | **FM Global Approved** | **Equipment Name** |
| F23-TR-601 | 5000 | 33/11 | Dyn11 | Sealed | KNAN | Yes | FIM III Booster Station 33/11kV Transformer |
| **Reference Documents** | | 2000-F23-SLD-EL-40001 – FIM III Booster Station HV Switchboard Single Line Diagram<br>2000-F23-SLD-EL-40002 – FIM III Booster Station HV Switchboard Key Interlocking Single Line Diagram | | | | | |
| **Reference Specifications** | | 2000-F00-STS-EL-10003 – Fluid Filled Distribution Transformers up to 20 MVA Specification<br>2000-F00-STS-EL-10004 – HV Electrical Testing Specification<br>2000-F00-STS-EL-10008 – Electrical Enclosures Specification<br>2000-F00-STS-GE-10002 – Site Specific Information<br>2000-F00-STS-GE-10001 – Packing and Shipping Specification<br>SPEC-00-G-05 – Protective Coating | | | | | |

**Scope of Supply**

**Work Included**

- Design, supply, and fabrication of One (1) Step down fluid insulated transformer including, but not limited to:
  - Transformer, fully assembled, FR3 ester insulated, for outdoor pad mounted installation
- The transformer complete with (as appropriate):
  - Off load tap changer
  - Sealed type
  - Radiators and valves (if applicable)
  - Instrumentation as appropriate
  - Marshalling panel/s
  - Wiring for instrumentation and controls including fan starters if required to comply with the environmental specification
  - Labelling and nameplate displaying all transformer data
  - Lifting lugs, as applicable
  - Any other items which are normally supplied with this equipment
- Manufacturer's standard Surface Protection equivalent to or better than Coating System CS1 in SPEC-00-G-05 – Protective Coating specification
- All ancillary items and special tools normally supplied with, or required to adjust and maintain all components associated with the equipment
- Laser engraved 316SS Stainless steel equipment nameplate
- All fasteners, gaskets, packers, shims and any other items necessary for site assembly
- Factory Acceptance Testing at manufacturer's premises
- Supply of all necessary documentation, drawings, and test documentation
- Packaging suitable for sea and road transport for delivery, including vibration monitoring during transportation
- Spare parts list
- Delivery to the delivery point.

**Work Excluded**

- Hold-down bolts, nuts, and washers at concrete foundations/structural steel
- Offboard cabling from Transformer to switchroom
- Offloading the equipment at vendor premises
- Installation and commissioning at site.

| | **Unit** | **Duty** | **Vendor** | **Notes** | **Rev** |
|---|---|---|---|---|---|
| **General** | | | | | |
| Transformer Electrical Ratings | kVA | 5000 | | 1 | A |
| Transformer Type | | Fluid insulated KNAN, Enhanced Protection with FM Global Approved with certificate | | 1 | B |
| Equipment Number and Name Plate | | Yes, on stainless steel plate | | 1 | A |
| Make | | Tyree Transformers | | 3 | A |
| Model | | 5000M6XX | | 3 | A |
| Country of Manufacture | | | | 2 | B |
| Compliance with Australian Standards | Comply/ Do Not Comply | Yes, AS2067, AS2374, AS60076, AS60214, AS60137 vendor to state compliance | | 1 | C |
| **Environmental Conditions** | | | | | |
| Plant Location | | Western Australia | | 1 | A |
| Service Location | Indoor/ Outdoor | Outdoor, direct sunlight and rain, polluted atmosphere | | 1 | A |

| | Unit | Duty | Vendor | Notes | Rev |
|---|---|---|---|---|---|
| Site Data | | Refer to 2000-F00-STS-GE-10002 – Site Specific Information | | 1 | A |
| **Physical Construction (Fluid Filled)** | | | | | |
| Tank Material and Radiator Material | | Tank – Mild Steel, painted PSL1 per AS2312; Radiators – Mild steel, hot dipped galvanised per AS4680 | | 3 | C |
| Mounting | | Pad Mount | | 1 | A |
| Sealed / Conservator | | Sealed with dry air cushion | | 3 | A |
| Cooling | | KNAN, FR3 Cargill, natural ester in sealed transformer | | 3 | A |
| Insulating Fluid Make | | Cargill FR3 (MIDEL en7131 option) | | 3 | A |
| Insulating Fluid Type | | K rated per FM3990; FR3 Cargill | | 3 | A |
| Comply with FM Global datasheet 3990 for construction for enhanced protection as per AS2067 Table 6.1 Note 1 | Comply/ Do Not Comply | | | 2, 5 | A |
| Minimum withstand pressure w/o Permanent Distortion | kPa | | | 2 | C |
| Minimum withstand pressure w/o rupture | kPa | | | 2 | C |
| Press. Relief Valve Fitted | Yes/No | Yes | | 3 | A |
| Press. Relief Valve Operating Pressure | kPa | | | 2 | C |
| Press. Relief Valve Flowrate (@103kPa) | SCFM | | | 2 | C |
| Cooling Fin Connection | Comply/ Do Not Comply | Cooling fins connected to the tank via piping flanges. | | 3 | A |
| Lifting Lugs | Yes / No | Yes | | 1 | A |
| Jacking Lugs | Yes / No | Yes | | 1 | A |
| Transformer Painting Specification | | PSL1 per AS2312 | | 3 | A |
| Transformer Paint Colour | | N42 Storm Grey per AS2700 | | 3 | A |
| HV Winding Conductor | | Copper | | 3 | A |
| LV Winding Conductor | | Copper | | 3 | A |
| Temperature Rating | | | | | |
| Rated Maximum Ambient Temperature | ºC | 50 | | 3 | A |
| Winding Temperature Rise | ºC | 65 | | 3 | A |
| Top Fluid Temperature | ºC | 60 | | 3 | A |
| Ability to Withstand Short Circuit in accordance with IEC 60076-5 | Yes / No | Yes | | 1 | A |
| Ability to withstand dynamic effects resulting from a primary or secondary short circuit | Yes / No | Yes | | 1 | A |
| Ability to withstand thermal effects resulting from a primary or secondary short circuit | Yes / No | Yes | | 1 | A |
| **Electrical Supply:** | | | | | |
| Primary Voltage | kV | 33 ± 10%, 3 Phase | | 1 | A |
| System Earthing, Primary Side | | 33kV solidly earthed at source, by GPPL | | 1 | A |
| Secondary Voltage (fully loaded) | kV | 11 ± 10%, 3 Phase, 4 wire – Neutral through NER | | 1 | A |
| System Earthing, Secondary Side | | 11kV NER – 25A/10s | | 1 | A |
| Vector Group | | Dyn11 | | 1 | A |
| Impedance | % | 7% | | 3 | A |
| Frequency | Hz | 50 ± 2% | | 1 | A |
| **33kV Winding Insulation** | | | | | |
| Ratings | kV rms | 36 | | 3 | A |
| Lightning-impulse withstand voltage (BIL) | kV peak | 200 | | 3 | A |
| Power-frequency withstand voltage | kV rms | 70 | | 3 | A |
| **11kV Winding Insulation** | | | | | |
| Ratings | kV rms | 12 | | 3 | A |
| Lightning-impulse withstand voltage (BIL) | kV peak | 95 | | 3 | A |
| Power-frequency withstand voltage | kV rms | 28 | | 3 | A |
| **Tap Changer** | | | | | |
| Type | | Off-load, Manual, provided with handled operable from outside tank | | 1 | A |
| Tapping Range | % | -5 to +5 | | 3 | A |

| | Unit | Duty | Vendor | Notes | Rev |
|---|---|---|---|---|---|
| Step size | % | 2.5 | | 3 | A |
| Location | | HV Winding | | 1 | A |
| 11kV Winding Star Point Neutral CT | | No | | 1 | A |
| **Cable Connections** | | | | | |
| Termination Details | | Porcelain bushings in cable box, bolted cable lugs. | | 3 | A |
| Cable boxes | | | | | |
| Gland Plate materials | | Aluminium 6mm thick | | 3 | A |
| Cable Entry | | Bottom Entry | | 1 | A |
| Degree of Protection | | ≥IP56 | | 3 | C |
| Covers | | Full removable, fully bolted, with earth stud. | | 1 | A |
| 33kV | | | | | |
| Number of cables | | 2 | | 1 | A |
| Insulation voltage | kV | 19 / 33 | | 1 | C |
| Type of cable | | 3C Cu XLPE/SWA/PVC | | 1 | B |
| Size of Conductor | $mm^2$ | 70 | | 1 | A |
| 11kV | | | | | |
| Number of cables | | 2 x 1C per phase in 2 trefoil groups | | 1 | A |
| Insulation voltage | kV | 6.35 / 11 | | 1 | C |
| Type of cable | | XLPE/SCR/PVC | | 1 | C |
| Size of Conductor | $mm^2$ | 240 | | 1 | A |
| **Equipment and Fittings** | | | | | |
| Main Tank | | | | | |
| Fluid Level indicator with Alarm Contacts | Yes / No | Yes | | 1 | A |
| Number of Contacts Required | | | | 2 | C |
| Pressure Relief Device with Trip contacts | Yes / No | Yes – Vendor to advise setpoint | | 3 | A |
| Number of Contacts Required | | | | 2 | C |
| Fluid Pressure Switch (Pressure Relief) | Yes / No | Yes – Vendor to advise details | | 3 | A |
| Number of Contacts Required | | | | 2 | C |
| Fluid Sampling Device | Yes / No | | | 2 | A |
| Number of Contacts Required | | | | 2 | C |
| Fluid Level indicator with Alarm Contacts | Yes / No | Yes | | 1 | A |
| Number of Contacts Required | | | | 2 | C |
| Fluid Thermometer Pocket | Yes / No | Yes | | 1 | A |
| Number of Contacts Required | | | | 2 | C |
| Earthing Terminal on Tank | Yes / No | Yes, 2 off | | 1 | A |
| Rating and Connection Diagram Plate | Yes / No | Yes, Stainless Steel | | 1 | A |
| Winding Temperature Indicator (RTD) with Alarm and Trip Contacts | Yes / No | Winding Temperature Thermostat Alarm Trip Voltage Free Contacts | | 3 | A |
| Fluid Temperature Indicator (OTI) with alarm and trip contacts | Yes / No | Fluid Temperature Thermostat Alarm Trip Voltage Free Contacts | | 3 | A |
| **Control Circuit** | | | | | |
| Marshalling Box / Control Panel | | 316 Stainless steel | | 1 | A |
| Trip and Alarm Voltage (protection DC voltage) | V | 48VDC | | 1 | A |
| Auxiliary voltage | | 415 / 240V AC | | 3 | C |
| Instrumentation Connections | | Screw Cage Terminals WDU2.5 or equivalent | | 3 | A |
| **Test Requirements** | | | | | |
| Factory Acceptance Test | | Yes, witnessed with test values included in report | | 1 | A |
| Type Tests per AS/NZS 60076 | | Yes, test certificates required | | 1, 6 | A |
| Standard Routine Tests per AS/NZS 60076 | | Yes, documentation required | | 1 | A |
| DGA and Moisture Content Testing | | Yes, after testing and prior to release. | | 1 | A |
| **Shipping and Delivery Information** | | | | | |
| Dimensions | mm | Length x Width x Height | | 2 | A |
| Quantity of Fluid | L | | | 2 | A |
| Weight of Fluid | kg | | | 2 | A |
| Total Transformer Weight including Fluid | kg | | | 2 | A |

| | Unit | Duty | Vendor | Notes | Rev |
|---|---|---|---|---|---|
| **Vendor Statement of Certification** | | | | | |
| The Vendor shall provide an FM Global approved transformer certificate | | | | 2 | A |

| Notes |
|---|
| 1. Fixed. The Vendor shall comply with this data, as a minimum. Vendor shall state their applicable ratings/data or compliance. |
| 2. Vendor to Advise. |
| 3. Preferred, Vendor to advise and provide reasoning for difference. |
| 4. Vendor to provide a certificate stating all fluids have been tested and found free of PCBs. |
| 5. Vendor to provide details of the enhanced protection provided. |
| 6. Type test certificates to be provided with tender documentation. |

**Vendor Drawing and Document Requirements List (VDDR)**

E – Electronic Copy H – Hard Copy MDR – Manufacture's Data Report

FAT – Factory Acceptance Testing IOM – Installation and Operating Manual

| VDDR | | | Required with Tender | Required after Award | | As Built Required |
|---|---|---|---|---|---|---|
| | | | Copies | Copies | Weeks | Copies |
| **A** | | **Production Schedule** | | | | |
| | A01 | Production Schedule / Bar Chart | 1E | 1E | 2 | |
| | A02 | Monthly Report | | Monthly | | |
| **B** | | **Registers and General Drawing Requirements** | | | | |
| | B01 | Drawing Schedule / Register (Drawings Expected) | | 1E | 2 | |
| | B02 | General Arrangement with Foundation Loads (in AutoCAD 2010 and PDF) | | 1E | 2 | 1E |
| | B03 | Cross-section | | | | |
| **C** | | **Mounting and Erection Drawings** | | | | |
| | C01 | Foundation and Mounting Details (Provided in B02) | | 1E | 4 | 1E |
| | C02 | Pipe Spools | | | | |
| | C03 | Outline Dimensions, Cut-out and Support Details (Provided in B02) | | 1E | 4 | 1E |
| | C04 | Match Marked Erection Drawings | | | | |
| | C05 | Motor GA (with Fixing and Shaft Details, and Overall Dimensions) | | | | |
| **D** | | **Electrical Drawings (in electronic form as appropriate)** | | | | |
| | D01 | Single Line Diagrams | | | | |
| | D02 | Schematic Drawings | | | | |
| | D03 | Terminations / Loop Diagrams | | 1E | 4 | 1E |
| | D04 | Panel Layout and Fabrication Details | | | | |
| | D05 | Nameplate | | 1E | 4 | 1E |
| | D06 | Cable Block Drawings | | | | |
| | D07 | Electrical and Instrumentation GAs | | | | |
| **E** | | **Process and Piping Drawings** | | | | |
| | E01 | Process Flow Sheets | | | | |
| | E02 | Process and Instrument Diagram | | | | |
| | E03 | Pneumatic or Hydraulic Diagram | | | | |
| | E04 | Lubrication Diagram | | | | |
| **F** | | **Technical Data** | | | | |
| | F01 | Equipment Information Schedule | 1E | | | |
| | F02 | Electrical Performance data sheet | | 1E | 6 | |
| | F03 | Equipment List | | 1E | 6 | 1E |
| | F04 | Vendor's Surface Protection Details | 1E | 1E | 6 | |
| | F05 | Lubrication Data Schedule | | | | |
| | F06 | Material Schedule | | | | |
| | F07 | Bolt Schedule | | | | |
| | F08 | Fan Performance Curves | | | | |
| | F09 | Pump Performance Curves (with Duty Points Marked) | | | | |
| | F10 | Cable Schedule | | | | |
| | F11 | Vendor Catalogue Information | 1E | | | |
| | F12 | Calculations | | | | |
| | F13 | Completed $_{\text{Engineer}}$ Datasheets | 1E | | | |
| | F14 | Motor Datasheets | | | | |
| | F15 | Electrical Equipment List | | | | |
| | F16 | Instrument List | | | | |

| VDDR | | | Required with Tender | Required after Award | | As Built Required |
|---|---|---|---|---|---|---|
| | | | Copies | Copies | Weeks | Copies |
| | F17 | I/O List | | 1E | 6 | |
| | F18 | Communications Addressing and Signal Mapping List | | | | |
| **G** | | **Inspection, Certification and MDR** | | | | |
| | G01 | Inspection and Test Plans | 1E | 1E | 4 | |
| | G02 | Inspection and Test Procedures (generic) | | | | |
| | G03 | Quality Plans and Procedures (generic) | | 1E | 4 | |
| | G04 | Manufacturer's Data Report (MDR) – Index / Contents | | 1E | 4 | |
| | G05 | Manufacturer's Data Report (MDR) | | 1E | Delivery +2 | |
| | G06 | Material Certificates | | In MDR | | |
| | G07 | Statutory Authority Certificates | | In MDR | | |
| | G08 | Test Data and Other Certificates | | In MDR | | |
| | G09 | Welding Procedure / Specification | | | | |
| | G10 | Non-Destructive Testing (NDT) Procedures | | | | |
| | G11 | Material Safety Data Sheets | | | | |
| | G12 | Commissioning Plan and Procedure | | 1E | Delivery - 4 | |
| | G13 | Structural Design Certification | | | | |
| | G14 | Performance Guarantee Statement | | | | |
| **H** | | **Shipping and Packing Details** | | | | |
| | H01 | Material / Shipping List | | 1E | Delivery -4 | |
| | H02 | Package Type and Quantity (Carton, Case, Crate) | | 1E | Delivery -4 | |
| | H03 | Shipping Packages – Weights and Dimensions | | 1E | Delivery -4 | |
| | H04 | Shipping and Handling Procedure | | 1E | 12 | |
| **J** | | **Manuals** | | | | |
| | J01 | Installation, Operation and Maintenance Manual (Preliminary) | | 1E | Delivery -4 | |
| | J02 | Installation, Operation and Maintenance Manual (Final) | | 1E | Delivery | |
| **K** | | **Spare Parts Information** | | | | |
| | K01 | Spare Parts Catalogue and Recommended Spares List (Copy also in Manuals) | 1E | 1E | 8 | |
| | K02 | Recommended Spare Parts List with Pricing (Not Included in Manuals) | 1E | 1E | Prelim 4<br>Delivery +1 | |
| | K03 | Special Tools list | 1E | 1E | 8 | |
| **M** | | **General** | | | | |
| | M01 | Control Philosophy and Logic Diagram | | | | |
| | M02 | Functional Description | | | | |